

Emotion detection from facial expressions using deep learning

Abstract

Face emotion recognition is essential for emotionally intelligent systems in education, healthcare, and human–computer interaction. However, most existing methods rely on convolution-based models that focus mainly on local facial features, are sensitive to lighting and pose variations, and operate without user identity integration, which limits personalization and increases system complexity due to separate processing models. To address these challenges, this project proposes a Vision Transformer (ViT) based multi-head deep learning framework that primarily focuses on accurate face emotion recognition while supporting face recognition as an additional feature. The ViT backbone divides facial images into patches and employs self-attention to learn global facial representations, enabling reliable detection of subtle and complex emotional patterns. The primary output head performs multi-class emotion classification, while a secondary output head generates identity embeddings for face recognition. This multi-task learning design enhances feature sharing, reduces redundancy, and improves computational efficiency without compromising emotion recognition accuracy. The proposed system is designed to operate efficiently in both image and video scenarios and supports real-time emotion analysis. Its modular architecture allows easy integration with surveillance and interactive platforms and enables future extensions such as emotion-based behavior analysis and personalized system responses. The system also facilitates future extensions such as behavior prediction and personalized recommendation based on emotional trends.

System Requirements

Hardware Requirements

- **Processor:** Minimum Intel i5 / AMD Ryzen 5 or equivalent
- **RAM:** 16 GB or higher
- **Storage:** 512 GB SSD (recommended for faster processing)

Software Requirements

- **Operating System:** Windows 10 / Windows 11
- **Development Platform:** Python 3.9 or above
- **Frameworks:** Flask for web dashboard
- **Database:** MySQL
- **AI & Machine Learning Tools:** OpenCV, Pillow, TensorFlow / PyTorch

- **Python Packages:** NumPy, Pandas, Scikit-learn, Matplotlib

Existing System

- **Traditional Machine Learning Approaches**

Early face emotion recognition systems used hand-crafted features such as Local Binary Patterns (LBP), Histogram of Oriented Gradients (HOG), and facial landmarks combined with classifiers like Support Vector Machines and K-Nearest Neighbors. These methods require manual feature design and perform poorly under variations in lighting, pose, and facial occlusions.

- **Convolutional Neural Network (CNN) Based Models**

CNN-based models automatically learn facial features from images and significantly improved emotion recognition accuracy. However, they mainly capture local spatial information and struggle to model long-range dependencies between different facial regions, which affects performance for subtle and complex expressions.

Drawbacks

- Limited global feature learning
- Sensitive to lighting and pose changes
- Low accuracy for subtle emotions
- High computational cost
- Separate models increase complexity
- Poor real-time performance
- Requires large datasets

Proposed System

The proposed system aims to improve the accuracy and robustness of face emotion recognition by adopting a transformer-based deep learning approach.

- **Vision Transformer Based Emotion Recognition**

The system utilizes a Vision Transformer (ViT) as the core feature extraction and learning algorithm. Facial images are divided into fixed-size patches and converted into embeddings, which are processed using self-attention mechanisms to capture global relationships among facial regions.

- **Multi-Head Face Analysis and System Integration**

A multi-head learning architecture is employed to support both emotion recognition and face recognition within a single unified model. While the main head predicts emotional states, a secondary head generates identity embeddings for optional face recognition.

Advantages

- Enables accurate understanding of human emotions in real time.
- Improves interaction between humans and intelligent systems.
- Supports personalized services based on emotional state.
- Helps in early identification of stress, fatigue, or negative emotions.
- Enhances decision-making in education, healthcare, and customer service.
- Reduces manual observation and human error in emotion analysis.
- Useful for behavioral studies and psychological analysis.
- Increases user engagement in interactive applications.

Module Description

1. Face Emotion Analyzer

The Face Emotion Analyzer is the core module responsible for integrating the front-end, backend, database, and deep learning components into a unified system. The module is developed using **Python** as the programming language and **Flask** as the web framework to handle routing, request processing, and system control. The user interface is designed using **HTML, CSS, and Bootstrap**, ensuring responsive and user-friendly interaction. **Wamp Server** is used to manage the local server environment and support the MySQL database connection. **MySQL** stores user details, emotion prediction results, identity information, timestamps, and system logs. Essential libraries such as OpenCV, NumPy, Pandas, and deep learning frameworks (TensorFlow/PyTorch) are used for image processing and model execution. This module acts as the central controller that manages image input, calls prediction models, processes outputs, and displays results through the web interface.

2. System User

The System User module manages all users interacting with the system and ensures secure access through authentication and authorization mechanisms. It controls user registration, login validation, session management, and access permissions. The module is divided into the following actors:

2.1 Administrator

The Administrator manages the overall system operations. This actor has full control over user accounts, including creating, updating, and deleting user records. The administrator can monitor system activity, manage datasets, review emotion prediction logs, and supervise system performance. The administrator also ensures security policies and access controls are properly maintained.

2.2 Registered User

The Registered User interacts directly with the system to analyze facial emotions. This actor can log in securely, upload images, or use live camera input for emotion detection. The registered user can view predicted emotion results, identity information (if enabled), and access their personal emotion history stored in the database.

3. Model Development

The Model Development module is responsible for designing, training, and optimizing the Vision Transformer (ViT) based multi-head deep learning model. The dataset containing labeled facial emotion images is preprocessed and used for training. The ViT model divides facial images into patches and uses self-attention mechanisms to capture global facial relationships. The model is trained to classify emotions accurately while also learning identity-related embeddings for face recognition. Performance evaluation metrics such as accuracy and loss are used to validate the model before deployment.

3.1 Data Collection and Preprocessing

This sub-module handles the acquisition of labeled facial emotion datasets. The collected images are cleaned, resized, normalized, and augmented to improve generalization. Face detection is applied to remove background noise and ensure that only relevant facial regions are used for training. The prepared dataset is then divided into training and testing sets.

3.2 Vision Transformer Architecture Design

This sub-module focuses on designing the ViT backbone model. The input facial images are divided into fixed-size patches, converted into embeddings, and enriched with positional encoding. Transformer encoder layers with self-attention mechanisms are configured to learn global relationships between facial regions. A multi-head output structure is defined to support both emotion classification and identity embedding generation.

3.3 Model Training and Optimization

In this stage, the ViT multi-head model is trained using the prepared dataset. The primary head is optimized for multi-class emotion classification, while the secondary head learns discriminative identity embeddings. Optimization techniques such as learning rate adjustment, regularization, and batch normalization are applied to improve convergence and prevent overfitting.

3.4 Model Evaluation and Validation

This sub-module evaluates the trained model using performance metrics such as accuracy, precision, recall, F1-score, and loss. Confusion matrices are analyzed to measure classification performance across emotion categories. The validated model is then saved and prepared for deployment in the prediction module.

4. Facial Emotion Predictor

This module performs real-time or image-based emotion classification. After face detection and preprocessing, the input image is passed to the trained Vision Transformer model. The primary output head of the model predicts the emotion category (happy, sad, angry, fear, surprise, disgust, neutral). The predicted result is sent to the Flask backend and displayed on the user interface. The emotion result is also stored in the database for analysis and reporting.

5. Face Identity Predictor

The Face Identity Predictor module works as an additional feature to identify the person whose emotion is being analyzed. The secondary head of the multi-head model generates identity embeddings, which are compared with stored embeddings in the database. If a match is found, the system retrieves and displays the corresponding user identity. This feature enables personalization and helps link emotional states with specific individuals for applications such as smart classrooms or behavioral analysis.

6. Result Visualization

The Result Visualization module presents emotion predictions and identity information in a clear and interactive format. Using Bootstrap-based dashboards, the system displays detected emotion labels, timestamps, and historical analysis. Graphs and charts can be generated to show

emotion trends over time, helping in behavioral monitoring and decision-making. This module enhances user understanding and provides meaningful insights from stored data.